A3 Assignment Report

A. Introduction

For this image classification project, we have decided to adopt an approach based on fine-tuning pre-trained models, such as ResNet, RegNet, and others. In total, we explored five different models. Initially, we attempted to use Weights and Biases for tracking experiments. However, given difficulties, we decided to restructure the code and reimplement it in a notebook format. Each notebook corresponds to a specific model, with its name and validation accuracy reflected in the filename. This approach allowed us to retain the accuracy scores and visualize the training and evaluation graphs for each model.

In this report, we will first present the data augmentation techniques applied to the training dataset to enhance model robustness. Then, we will detail the hyperparameters used across the five models evaluated. Finally, we will focus on the best-performing model, outlining the personalized parameter adjustments made to optimize its performance.

B. Exploration

B.1. Data augmentation

To enhance the training data and improve model generalization, we applied a series of transformations. The transformations included resizing to 224×224, random horizontal and vertical flips, random rotations up to 15°, and color jittering to vary brightness, contrast, saturation, and hue. Additionally, a random resized crop ensured scale diversity, followed by normalization.

B.2. Hyperparameters

For training, each model was fine-tuned over 10 to 15 epochs with a batch size of 64. The learning rate was set to 0.005, and the optimization utilized Stochastic Gradient Descent (SGD) with a momentum of 0.9. These hyperparameters provided a balance between training speed and stability, ensuring convergence across all models.

B.3. Model and accuracies

The five models we used in this project were ResNet-50 [1], ResNet-152 [1], DenseNet [2], RegNetY-16GF [4], and the Swin Transformer [3]. With the data transformations and hyperparameters settings described above, these models achieved the following validation accuracies: 82.72% for ResNet-50, 82.84% for ResNet-152, 82.92% for DenseNet, 85.84% for RegNetY-16GF, and 87.60% for the Swin Transformer. We can see all the accuracy plots in the appendix and graphics section.

From these results, the Swin Transformer is by far the best-performing model. Therefore, we decided to focus on this model and explore ways to improve its performance.

C. Swin Transformer tuning

In order to enhance the performance of the Swin Transformer, we first changed the optimizer from SGD to AdamW and lowered the learning rate to 0.0001. Additionally, we introduced a cosine scheduler with 10 warm-up steps. These adjustments increased the validation accuracy to **89.32%**. On the competition's test set, this configuration achieved **89.4%**.

Then, we experimented with adding a dropout layer of 0.5 to reduce overfitting, but it decreased the validation accuracy to **88%**. Adjusting the dropout rate to 0.3 and increasing the learning rate to 0.0003 improved the results to **88.52%** on validation and **89.9%** on the test set.

We also modified the data augmentation strategy to be more aggressive, particularly by increasing the variability in crop resizing and rotation. We also experimented with changing the batch size to 32 and 128, but it had no impact on the results.

Thanks to all these changes, we observed a reduction in overfitting: the training accuracy peaked at **93%** with dropout, compared to over **95%** without it. This balance suggests that the inclusion of dropout helped the model generalize better, despite slightly lower training performance.

Table 1. Validation accuracy (%) for different models and configurations

Configuration	ResNet-50	ResNet-152	DenseNet	RegNetY-16GF	Swin Transformer
SGD	82.72	82.84	82.92	85.84	87.60
AdamW + Scheduler	-	-	-	-	89.32
AdamW + Scheduler + Dropout 0.5	-	-	-	-	88.00
AdamW + Scheduler + Dropout 0.3	-	-	-	-	88.52

D. Limitations and possible future work

One of the main limitations of this project is the computational power available, as we are constrained by the GPUs used for training. With access to more powerful hardware, we could explore more complex models, such as those from the DINOv2 family. Despite several attempts to train these models, their training process was significantly slower, which led us to prioritize testing other architectures.

A more detailed study of the impact of different data augmentation techniques on this specific dataset might be of interest. Identifying the most effective augmentations might improve the model's performance by a few points, while optimizing accuracy. It is definitely an area for future exploration!

E. Author

Eliott Vigier

Appendix and graphics

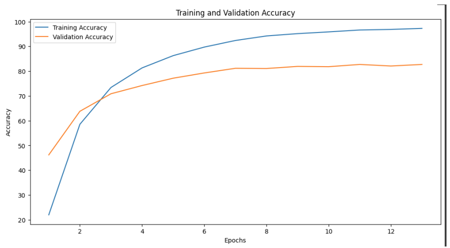

Figure 1. The curves show the evolution of training and validation accuracy for the ResNet-50 model. The maximum validation accuracy achieved is **82.72%**.

Figure 2. The curves illustrate the training and validation accuracy for the ResNet-152 model. The maximum validation accuracy achieved is **82.84%**.

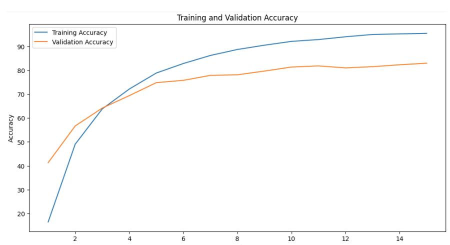

Figure 3. The curves represent the training and validation accuracy for the DenseNet model. The maximum validation accuracy achieved is **82.92%**.

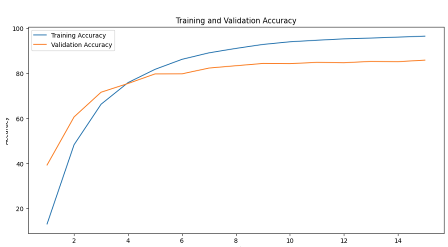

Figure 4. The curves show the training and validation accuracy for the RegNetY-16GF model. The maximum validation accuracy achieved is **85.84%**.

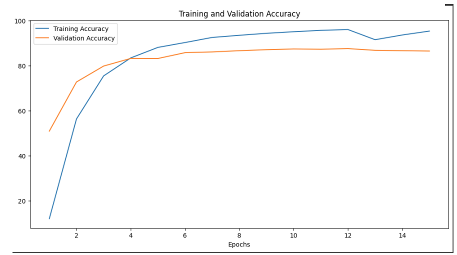

Figure 5. The curves display the training and validation accuracy for the Swin Transformer model with SGD optimizer. The maximum validation accuracy achieved is **87.60%**.

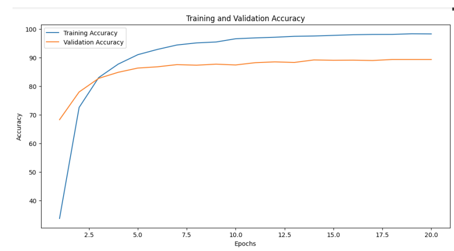

Figure 6. The curves display the training and validation accuracy for the Swin Transformer model with AdamW optimizer. The maximum validation accuracy achieved is **89.32%**.

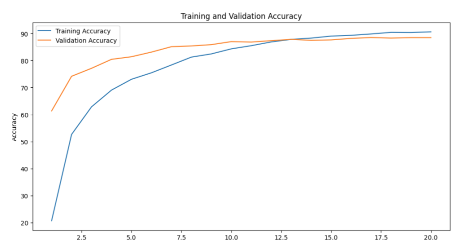

Figure 7. The curves display the training and validation accuracy for the Swin Transformer model with AdamW optimizer and 0.3 dropout. The maximum validation accuracy achieved is **88.52%** and **89.94%** on the test.

References

[1] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 1

[2] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *CVPR*, 2017. 1

[3] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, 2021. 1

[4] Ilija Radosavovic, Raj Prateek Kosaraju, Ross B. Girshick, Kaiming He, and Piotr Dollár. Designing network design spaces. In *CVPR*, 2020. 1